SULIT

SECP2523 DATABASE

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ALTERNATIVE ASSESSMENT REPORT: PHASE 1

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Case Study	Reporting System
System Name	KADA
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SECTION A (SYSTEM'S OVERVIEW)

1.1 Overall Description

The current reporting system in KADA is a manual and paper-based process that handles the preparation and distribution of financial and operational reports. This system relies heavily on employees manually collecting data from various records, compiling it into reports, and distributing them to members or decision-makers.

Member Financial Reports
Financial statements are manually compiled by retrieving data from various physical records. After preparation, statements are distributed to members, who must review, sign, and return them for verification.

Operational Reports
The organization lacks an efficient mechanism to generate real-time organizational summaries, such as loan approval rates, member contribution trends, or financial performance. These reports are created manually, requiring significant effort and time, and are often outdated by the time they are completed.

Data Storage and Retrieval
The paper-based storage of reports and records makes it difficult to retrieve data for analysis or auditing purposes. This lack of quick access to data hinders effective decision-making and reduces operational agility.

In a nutshell, the reporting system of KADA rely on manual processes and physical documentation which have significantly limits KADA's efficiency, accuracy and ability to generate insights for decision-making.

1.2 Project Weakness or Improvement

The current reporting system at KADA suffers from multiple critical weaknesses due to its reliance on manual processes and scattered data storage.

Fragmented Data Storage

Member information, loan records, and financial data are stored in separate physical files and documents, resulting in a lack of integration. This fragmentation makes it difficult to integrate data for comprehensive reporting. A database would resolve this by storing all the data in a centralized and relational structure, allowing for efficient data management and retrieval.

Manual and Time-Consuming Reporting

Reports are compiled manually by collecting data from various records, which is both time-consuming and error prone. The repetitive data collecting and formatting tasks by clerks may increase the risk of human errors. With a database, automated reporting processes could be implemented using SQL queries and predefined report templates to ensure accuracy and save time.

Lack of Real-Time Reporting Capabilities

The current system cannot generate real-time reports because data updates and report creation are performed manually at scheduled intervals. This delay in reporting hinders timely decision-making. A database would enable real-time updates, where reports can be generated dynamically based on the most recent data.

No advanced Data Analysis

The current system does not support advanced data analysis or summary reporting, limiting the ability to identify trends or insights. A database system could support features like summary tables, dashboards, and analytics tools to improve data-driven decision-making.

These weaknesses highlight the limitations of a manual system. A database-centric approach is necessary to address these issues, providing a streamlined, accurate, and accessible solution for reporting.

1.3 Proposed Module

To overcome the weaknesses outlined in the current reporting system, a digitalize system with comprehensive database-driven reporting module is proposed. This module aims to streamline the reporting process, ensure data accuracy, and enhance real-time accessibility.

Centralized Data Storage

A relational database will be implemented to store all member information, loan records, and financial data in single, centralized system. This integration ensures that data is easily accessible and eliminates the need to search through fragmented files or documents.

Automated Reporting

The module will include automated reporting capabilities powered by SQL queries and predefined templates. Reports such as financial summaries, loan performance, and member contribution details can be generated. This reduces manual effort, saves time, and minimizes errors in data handling.

Real-Time Report Generation

Real-time updates will be supported through the database, allowing users to access up-to-date reports at any time. This feature ensures that decision-makers have the latest information when analyzing financial or operational trends, improving responsiveness and efficiency.

Data Analytics and Visualization

Advanced analytics tools will be integrated to enable trend analysis and summary reporting. Dashboards and visualized reports will be provided to offer a clear and concise view of key metrics, such as member registration and loan application trends or member contribution over time.

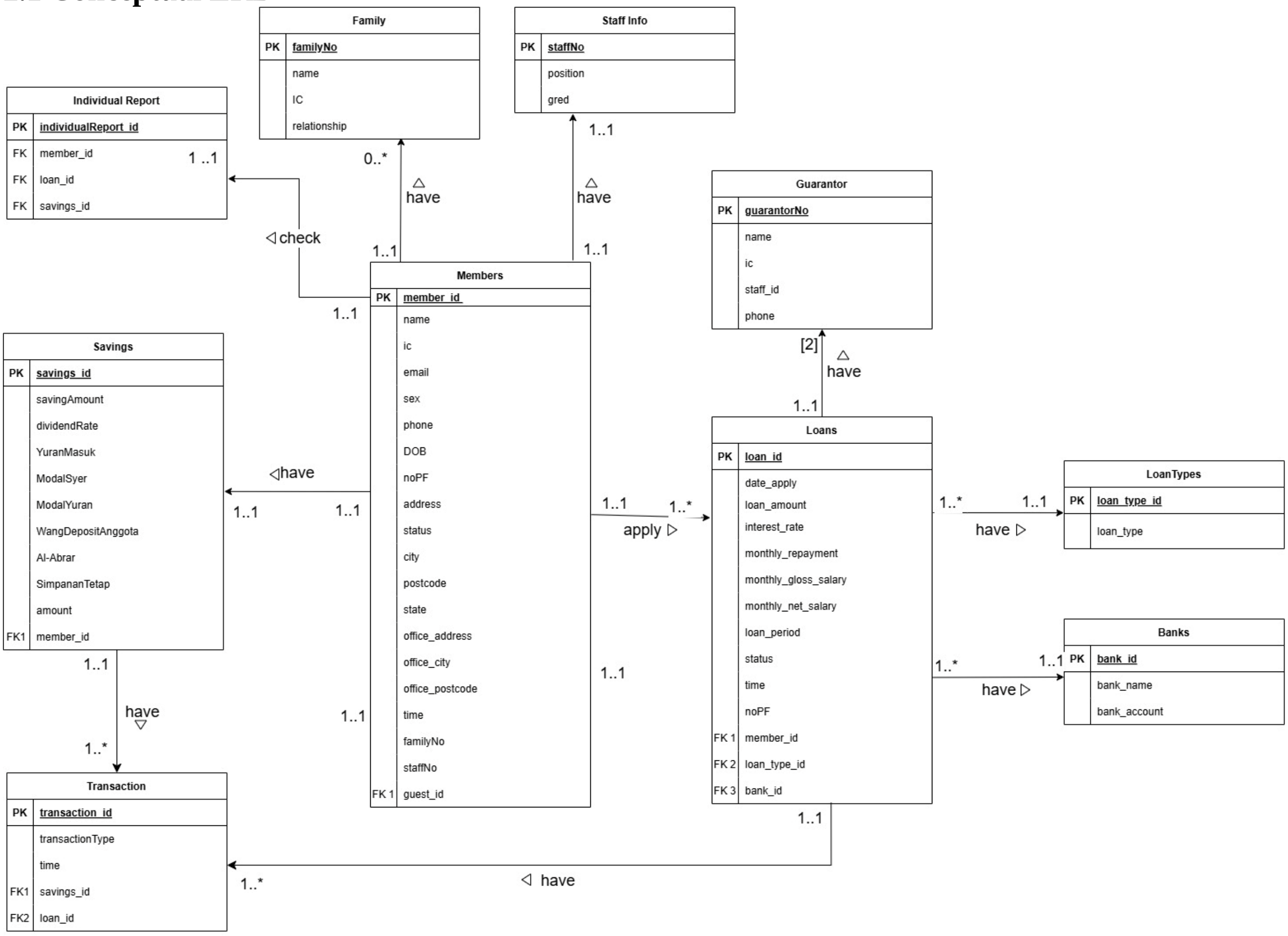
User-Friendly Access

The proposed module will feature a user-friendly interface that allows clerk to easily retrieve, filter, and export reports. Access controls and permissions will also be implemented to ensure data security and limit unauthorized access ton sensitive information.

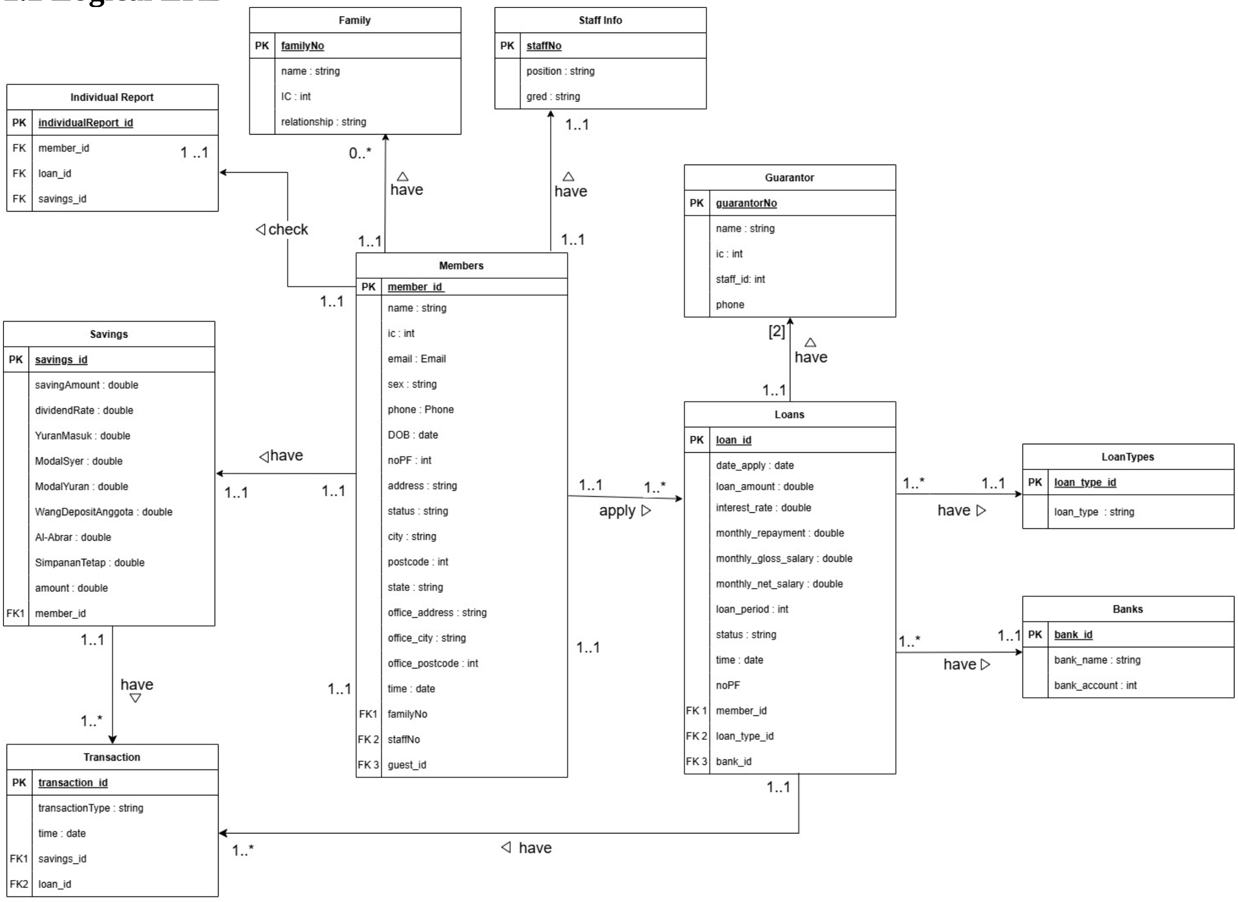
This reporting module addresses the core weaknesses of the current system by leveraging database technology to provide an efficient, accuratem and accessible solution for KADA’s reporting needs.

SECTION B (DATABASE PLANNING AND DESIGN)

2.1 Conceptual ERD



2.2 Logical ERD



2.3Data Dictionary for Logical ERD

2.3.1 Identifying Entity

Entity Name	Description	Aliases	Occurrence
Member	General term describing all user logged in as member using KADA-eServe	Registered staff	Occurs when a guest registration application approved by admin
Transaction	General term describing all transaction record key-in by admin	Deposit amount, loan repayment	Occurs when admin key-in the amount of deposit or loan repayment for each member
Savings	General term describing the savings info of each member	Yuran, Modal, Deposit, Simpanan	Occurs when a guest registration application approved by admin
Loan	General term describing the loan info of member	Pembiayaan	Occurs when a member submits the loan application form to system
Individual Report	General term describing the individual report of each member saving and loan info.	Personal report	Occurs when a guest registration application approved by admin
Family	General term describing the family info of each member	Pewaris, Keluarga	Occurs when a guest registration application approved by admin
Staff Info	General term describing the staff position, grade info	KADA staff Info	Occurs when a guest registration application approved by admin
Guarantor	General term describing the guarantor of the member	Warrantor	Occurs when a guest registration application approved by admin
LoanTypes	General term describing the loan info of member	Loan type	Occurs when a member submits the loan application form to system
Banks	General term describing the bank account info of member	Bank account	Occurs when a member submits the loan application form to system

2.3.2 Identifying Relationship Types

Entity name	Multiplicity	Relationship	Entity Name	Multiplicity
Member	1..1	have	Savings	1..1
	1..1	have	Family	0.. *
	1..1	have	Staff Info	1..1
	1..1	apply	Loans	1.. *
	1..1	check	Individual Report	1..1
Loan	1..1	have	Transaction	1.. *
	1..1	have	Guarantors	[2]

	1..1	have	LoanTypes	1..*
	1..1	have	Banks	1..1
Savings	1..1	have	Transaction	1.. *

2.3.3 Attributes Description

Entity Name	Attributes	Data Type & Length	Description	Nulls	Multi-valued	Example
Members	member_id (PK)	int	Uniquely identifies each member	No	No	5
	name	varchar(255)	Members' full name	No	No	Ahmad
	ic	int	Members' identification card number	No	No	981119-01-8732
	email	varchar(255)	Account email for receive notification	No	No	ahmad@gmail.com
	sex	varchar(255)	Members' gender	No	No	Lelaki
	phone	varchar(255)	Members' phone number	No	No	011-16596391
	DOB	date	Members' birth date	No	No	19/11/1998
	noPF	int	Members' personal profile	No	No	6
	address	text	Members' address	No	No	No 128 Taman U
	status	ENUM	Indicates the application state	No	No	Johor
	city	text	City where the member is located	No	No	Muar
	postcode	int	Postcode where the member is located	No	No	84000
	state	text	State where the member is located	No	No	Kelantan
	office_address	text	Office address of member	No	No	Peti Surat 127
	office_city	text	Office city of member	No	No	Bandar Kota Bharu
	office_postcode	int	Office postcode of member	No	No	15710
	time	date	The date of submission application	No	No	05/1/2025

	familyNo	int	Family record of members	Yes	Yes	2
	staffNo	int	Staff number of member	No	No	4672
Transaction	Transaction_id (PK)	int	Uniquely identified transaction record	No	No	6
	transactionType	ENUM	Type of transaction made	No	Yes	Loan repay , Deposit
	time	date	Date of transaction made	No	No	19/1/2025
	Savings_id (FK)	int	Indicated to the specific savings account info	No	No	7
Savings	savings_id (PK)	int	Uniquely identified savings record	No	No	56
	savingAmount	float	Amount of the member savings	No	No	3000
	dividendRate	float	Yearly dividen rate	No	No	0.03
	YuranMasuk	float	Amount of members' Yuran Masuk	No	No	2550
	ModalSyer	float	Amount of members' Modal Syer	No	No	1480
	ModalYuran	float	Amount of members' Modal Yuran	No	No	1320
	WangDepositAnggota	float	Amount of members' deposite	No	No	2550.50
	Al-Abrar	float	Amount of members' Al-Albrar	No	No	1000.50
	SimpananTetap	float	Amount of members' fixed deposit	No	No	6000.50
	amount	float	Total amount of the savings	No	No	20000
	member_id (FK1)	int	Indicated to the member who own the savings record	No	No	6
Family	FamilyNo (PK)	int	Uniquely identified family record of member	No	No	4
	name	varchar(255)	Members' family	No	No	Ismail

			name			
	IC	varchar(255)	Members' family ic	No	No	931118-01-1236
	relationship	varchar(255)	Relationship between the family and member	No	No	Adik lelaki
Staff Info	staffNo	int	Uniquely identified each staff	No	No	9824
	position	varchar(255)	Staffs' position in KADA	No	No	perkhidmatan pelanggan
	gred	varchar(255)	Staffs' grade in KADA	No	No	M40
Guarantor	GuarantorNo (PK)	int	Uniquely identified guarantor of member	No	No	7
	name	varchar(255)	Guarantors' name	No	No	Siti
	ic	varchar(255)	Guarantors' IC	No	No	991231-01-9823
	staff_id	int	Guarantors' staff ID in KADA	No	No	7
	phone	varchar(255)	Guarantors' phone number	No	No	019-7113981
Loans	loan_id (PK)	int	Uniquely identified each application record	No	No	6
	date_apply	date	Submission date of application	No	No	20/1/2025
	loan_amount	float	Amount of loan to apply	No	No	2000.50
	interest_rate	float	Interest rate for loan	No	No	0.05
	monthly_repayment	float	Monthly repayment of member	No	No	200
	monthly_gloss_salary	float	Monthly gloss salary of member	No	No	4000
	monthly_net_salary	float	Monthly net salary of member	No	No	3500
	loan_period	date	Period of loan apply	No	No	18/2/2028
	status	ENUM	Status of the application	No	No	approved
	time	date	Submission date of the application	No	No	19/1/2025
	noPF	int	Indicated to the	No	No	4

			personal profile of member			
	member_id (FK 1)	int	Indicated to member ID	No	No	8
	loan_type_id (FK 2)	int	indicated to type of loan member apply	No	No	4
	bank_id (FK 3)	int	Indicated to bank record of member	No	No	6
LoanTypes	loan_type_id (PK)	int	Uniquely identified the type of loan	No	No	5
	loan_type	varchar(255)	Type of loan to be applied	No	No	Simpanan Tetap
Banks	bank_id (PK)	int	Uniquely identified each members' bank info	No	No	8
	bank_name	varchar(255)	Members' bank name	No	No	Maybank
	bank_account	int	Members' bank account number	No	No	123456789000
Individual Report	individualReport_id (PK)	int	Uniquely identified each members' individual report	No	No	8
	member_id (FK)	int	Indicated to which member own the report	No	No	7
	loan_id (FK)	int	Indicated to the loan application that the member made	No	No	2
	savings_id (FK)	int	Indicated to the savings record of the member	No	No	5

2.4 Relational Database Schema

The Relational Database Schema represents the tables derived from the Logical ERD, showing the primary keys, foreign keys and relationships. Below represent the relational database schema for this Reporting Module:

1. Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city,postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id)

   Primary Key: member_id

   Foreign Key: guest_id references Guest(guest_id), familyNo references Family(familyNo), staffNo references StaffInfo(staffNo)

2. Staff Info (staffNo, position, gred)

   Primary Key: staffNo

3. Family (familyNo, name, IC, relationship)

   Primary Key: familyNo

4. Loan(loan_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gloss_salary, monthly_net_salary, loan_period, status, time, noPF, member_id, GuarantorNo, loan_type_id, bank_id)

   Primary Key: loan_id

   Foreign Key: member_id reference Member(member_id), GuaratorNo reference Guarantor(GuarantorNo), loan_type_id reference Loan Types(loan_type_id), bank_id reference Bank(bank_id)

5. Guarantor(GuarantorNo, name, ic, staff_id, phone)

   Primary Key: GuarantorNo

6. Loan Types(loan_type_id, loan_type)

   Primary Key: loan_type_id

7. Banks(bank_id, bank_name, bank_account)

   Primary Key: bank_id

8. Savings (savings_id, savingAmount, dividendRate, YuranMasuk, ModalSyer, ModalYuran, Wang DepositAnggota, Al-Abrar, SimpananTetap, amount, member_id)

Primary Key: savings_id

Foreign Key: member_id reference Member(member_id)

9. Individual Report (individualReport_id, member_id, loan_id, savings_id)

   Primary Key: individualReport_id

   Foreign Key: member_id reference Member(member_id), loan_id reference Loan(loan_id), savings_id reference Savings(savings_id)

10. Transaction(transaction_id, transactionType, time, savings_id, loan_id)

    Primary Key: transaction_id

    Foreign Key: loan_id reference Loan(loan_id), savings_id reference Savings(savings_id)

2.5 Normalization
Member Table
1NF
MemberForm (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city,postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id, familyNo, name, IC, relationship, staffNo, position, gred, savings_id, savingAmount, dividendRate, YuranMasuk, ModalSyer, ModalYuran, Wang DepositAnggota, Al-Abrar, SimpananTetap, amount)
Primary Key: member_id

2NF
Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city,postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id, familyNo, name, IC, relationship, staffNo, position, gred)
Primary Key: member_id

Savings (savings_id, savingAmount, dividendRate, YuranMasuk, ModalSyer, ModalYuran, Wang DepositAnggota, Al-Abrar, SimpananTetap, amount, member_id)
Primary Key: savings_id
Foreign Key: member_id reference Member(member_id)

3NF
Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city,postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id)
Primary Key: member_id
Foreign Key: guest_id references Guest(guest_id), familyNo references Family(familyNo), staffNo references StaffInfo(staffNo)

Savings (savings_id, savingAmount, dividendRate, YuranMasuk, ModalSyer, ModalYuran, Wang DepositAnggota, Al-Abrar, SimpananTetap, amount, member_id)
Primary Key: savings_id
Foreign Key: member_id reference Member(member_id)

Family (familyNo, name, IC, relationship)
Primary Key: familyNo

Staff Info (staffNo, position, gred)
Primary Key: staffNo

Loan Table
1NF
LoanForm(loan_id, date_apply, loan_type, loan_amount, interest_rate, monthly_repayment, monthly_gloss_salary, monthly_net_salary, loan_period, status, time, noPF, member_id, GuarantorNo, name, ic, staff_id, phone, bank_name, bank_account)
Primary Key: loan_id

2NF
Loan(loan_id, date_apply, loan_type, loan_amount, interest_rate, monthly_repayment, monthly_gloss_salary, monthly_net_salary, loan_period, status, time, noPF, member_id, phone, bank_name, bank_account)
Primary Key: loan_id
Foreign Key: GuarantorNo reference Guarantor(GuarantorNo)

Guarantors(GuarantorNo, name, ic, staff_id)
Primary Key: GuarantorNo

3NF
Loan(loan_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gloss_salary, monthly_net_salary, loan_period, status, time, noPF, member_id, GuarantorNo, loan_type_id, bank_id)
Primary Key: loan_id
Foreign Key: member_id reference Member(member_id), GuaratorNo reference Guarantor(GuarantorNo), loan_type_id reference Loan Types(loan_type_id), bank_id reference Bank(bank_id)

Guarantor(GuarantorNo, name, ic, staff_id, phone)
Primary Key: GuarantorNo

Loan Types(loan_type_id, loan_type)
Primary Key: loan_type_id

Banks(bank_id, bank_name, bank_account)
Primary Key: bank_id